

# Project #6: GAME-BASED UPPER-LIMB ROBOTIC EXOSKELETON

## Adaptive Neurorehabilitation, Smart Assessment and Continuous Progress Monitoring

### 1. Project Description and Motivation

This graduation project will redesign and extend the previous unilateral upper-limb rehabilitation robot into a smaller, more compact and easier-to-use exoskeleton for supervised clinical sessions and future home-based rehabilitation. The system will support repetitive, task-oriented training of the shoulder, elbow and forearm/wrist while measuring joint motion, interaction force, applied torque and patient performance.

A monitor positioned in front of the patient will deliver motivating rehabilitation games and display score, level, timer, targets, progress and encouraging feedback. The game software will follow performance in real time and adapt both task difficulty and robotic assistance according to the patient's measured ability and progress.

### 2. Main Mechanical Redesign

- Shorten links, reduce unnecessary brackets and minimize the overall frame footprint.
- Relocate heavy components and use compact servo/BLDC actuators, low-backlash transmissions and lightweight materials.
- Improve cable routing and simplify alignment for different arm dimensions.
- Use adjustable cuffs, quick-release straps, mechanical stops and an accessible emergency stop.
- Adopt modular shoulder, elbow and forearm/wrist joints for easier testing, maintenance and future upgrades.

### Project Aim

To create a safe, engaging and data-driven rehabilitation platform that combines intelligent sensing, multimode adaptive control, game-based therapy, objective assessment and remote progress monitoring.

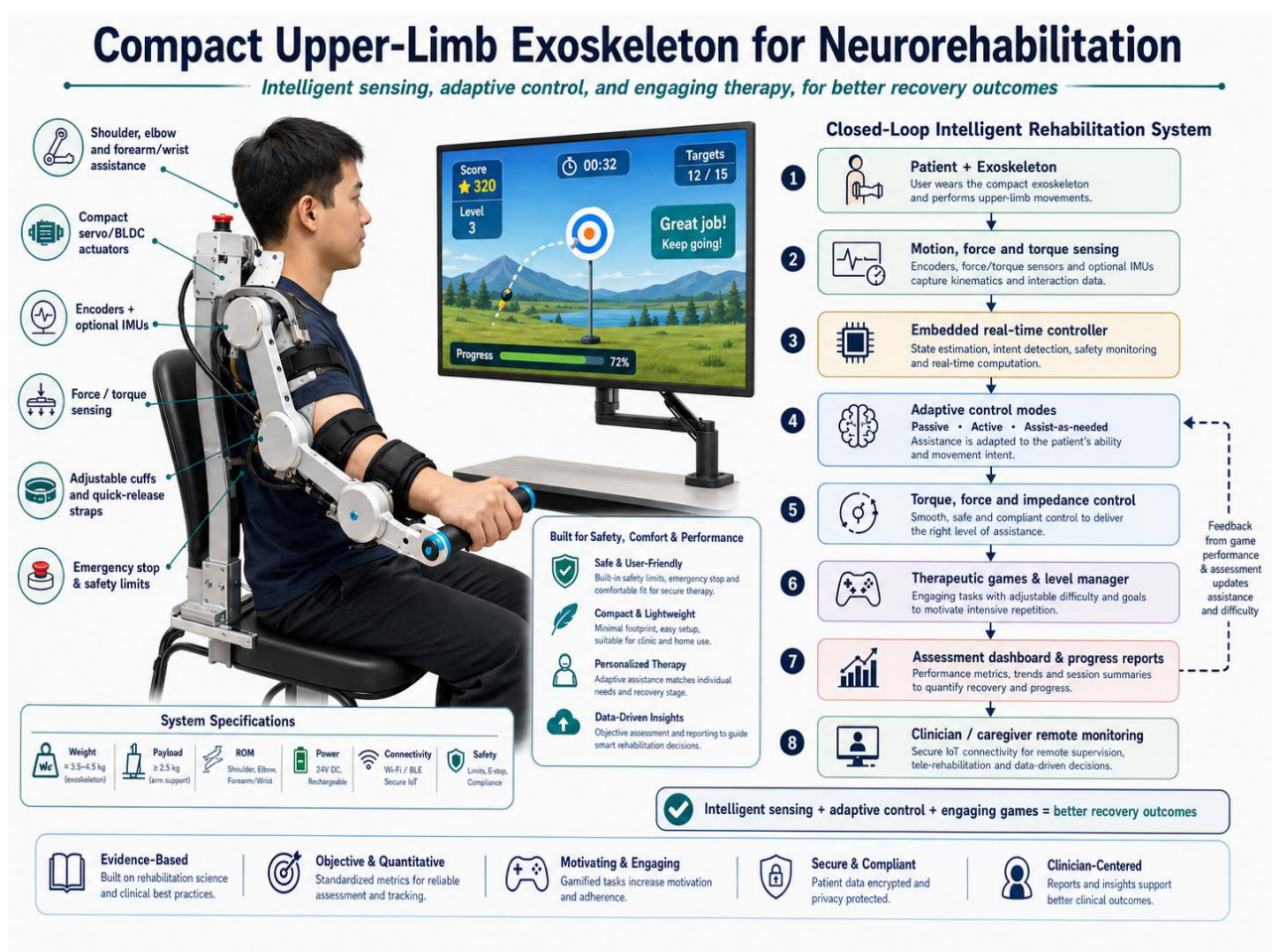


Figure 1. Compact exoskeleton, game monitor and closed-loop system concept.

# SYSTEM HARDWARE, SENSING AND ADAPTIVE CONTROL

## Hierarchical Control Architecture for Safe Adaptive Rehabilitation

*Intelligent sensing. Adaptive control. Engaging therapy. Better outcomes.*

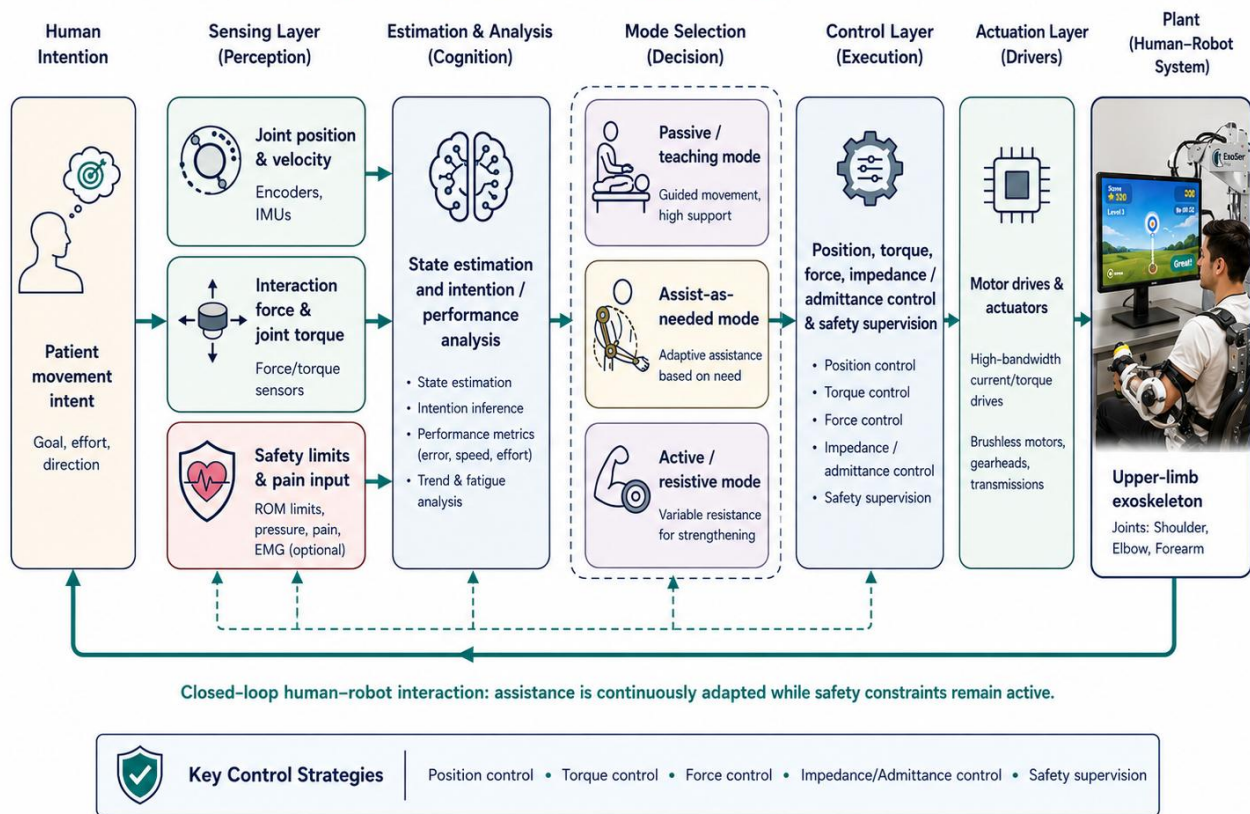


Figure 2. Hierarchical control architecture for safe adaptive rehabilitation

### 3. Hardware and Sensor Layout

**Mechanical structure:** Adjustable shoulder-elbow links, arm/forearm cuffs, quick release, mechanical stops and simplified compact support.

**Actuation:** Compact servo/BLDC motors with low-backlash transmissions and brakes where required.

**Motion sensing:** Absolute/incremental encoders and optional IMUs for joint position, velocity, range of motion and movement quality.

**Interaction sensing:** Joint torque sensors or motor-current estimation, plus an end-effector/load-cell force sensor.

**Embedded control:** Real-time MCU or industrial PC, motor drives, isolated I/O, watchdog, emergency stop and safe-stop logic.

**User interface:** Game monitor, clinician settings, calibration tools, patient start/stop controls, data logging and reporting.

### 4. Control Modes and Algorithms

- Passive / continuous passive motion: prescribed safe trajectories with adjustable speed, range and repetitions.
- Active mode: voluntary movement with minimal assistance while motion quality and effort are recorded.
- Assist-as-needed mode: only the missing effort is supplied; assistance decreases as performance improves.
- Resistive / strength mode: controlled opposing torque or viscosity-like resistance for advanced training.
- Teaching mode: therapist-guided trajectories can be stored and replayed as patient-specific exercises.

The low-level controller will combine position control with torque and force regulation. Impedance/admittance control will define a programmable spring-damper relationship between motion error and interaction force so that the robot remains compliant and comfortable. A supervisory controller will select gains, permitted ROM, assistance level and game difficulty using calibration data, safety rules and patient performance.

#### Safety Functions

Emergency stop • software joint limits • mechanical stops • force/torque thresholds • speed limits • watchdog • controlled safe stop

# GAME-BASED THERAPY, SMART ASSESSMENT AND PROJECT OUTPUTS

## Game-Based Therapy and Smart Patient Assessment Software

Adaptive • Data-Driven • Safe • Engaging



Figure 3. Game-based therapy dashboard, assessment metrics and adaptive level progression.

### 5. Therapeutic Games and Automatic Progression

The software will include point-to-point reaching, target interception, path tracking and object-transport exercises. Adjustable variables will include target size and distance, movement direction and range, permitted time, path complexity, assistance, resistance and repetition count. A baseline calibration session will define the initial level and safe workspace for each patient.

After every trial, the software will evaluate ROM, tracking error, accuracy, completion time, smoothness, interaction force, effort and successful repetitions. When predefined performance and safety criteria are satisfied, the patient advances to the next level. If performance declines, the system may repeat or simplify the task, enlarge targets, reduce range, extend time or increase assistance.

#### Assessment and Monitoring Software

- Patient profile, calibration and session plan
- Real-time joint angle, torque and force plots
- ROM, error, smoothness, success rate and effort
- Assistance and game-difficulty history
- Session comparison and exportable progress reports
- Optional secure IoT synchronization for clinician review

### 6. Continuation of the Previous-Year Project

Previous prototype video: <https://www.youtube.com/watch?v=UozokDtQbml>

The new team will review the existing mechanism, electronics, control code and test results, retain successful modules where appropriate, and implement the compact redesign and adaptive software described above.

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